



C# syntax cheat sheet

LearnHub course reference: C# Programming Fundamentals

Types you will use most

Type	Example	Use for
int	int lessons = 12;	Whole numbers
decimal	decimal fee = 1250.50m;	Money, exact values
double	double ratio = 0.75;	Measurements
bool	bool passed = true;	Yes or no
char	char grade = 'A';	One character
string	string name = "Aisyah";	Text

Decisions

```
if (score >= 70)
{
    Console.WriteLine("Passed");
}
else if (score >= 50)
{
    Console.WriteLine("Almost there");
}
else
{
    Console.WriteLine("Keep practising");
}

string grade = score switch
{
    >= 85 => "Distinction",
    >= 70 => "Pass",
    _ => "Not yet"
};
```

Loops

```
for (int i = 1; i <= 3; i++) {
    Console.WriteLine(i);
}

foreach (string lesson in lessons)
{
    Console.WriteLine(lesson);
}

while (attempts < 3) { attempts++; }
```

Methods

```
static decimal Average(int[] scores)
{
    if (scores.Length == 0) return 0;
    return (decimal)scores.Sum() /
    scores.Length;
}

decimal result = Average([72, 58, 90]);
```

Classes and properties

```
public class Course
{
    public Course(string title) => Title =
    title;

    public string Title { get; }
    public bool IsPublished { get; set; }
}

var course = new Course("C# Fundamentals");
course.IsPublished = true;
```

Collections

```
int[] scores = [72, 58, 90];
List<string> names = ["Aisyah", "Daniel"];
names.Add("Priya");

var codes = new Dictionary<string, string>
{
    ["WAPP"] = "Web Applications"
};
```

Strings and input

```
string message = $"Hello, {name}!";

if (int.TryParse(Console.ReadLine(), out int
hours))
{
    Console.WriteLine($"{hours} hours
logged");
}
```

.NET CLI

dotnet new console -o App	Create a project
dotnet build	Compile

Operators

+ - * / %	Arithmetic (7 / 2 is 3 with int)
== != < > <= >=	Comparison
&& !	And, or, not
??	Use a fallback when a value is null

dotnet run	Build and run
dotnet watch	Rebuild on save
dotnet test	Run unit tests

LearnHub course reference. Examples use C# 12 or later on .NET 10.